

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

STATION:	SALEM		
SYSTEM:	Reactor Protection System		
TASK:	Withdraw control rods to criticality, and TCAF Failure of Source Range Hi Flux Trip to BLOCK.		
TASK NUMBER:	N1120170101		
JPM NUMBER:	13-01 NRC Sim g		
ALTERNATE PATH:	☒	K/A NUMBER:	012 A4.03
APPLICABILITY:		IMPORTANCE FACTOR:	
	EO ☐		3.6 RO
	RO ☒		3.6 SRO
	STA ☐		
	SRO ☒		
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-IO.ZZ-0003, Rev. 39 (Checked 10/7/14)		
TOOLS AND EQUIPMENT:			
VALIDATED JPM COMPLETION TIME:	20 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	7/8/14
Validated By:	C Omlor SME or Instructor	Date:	10/15/14
Approved By:	 Training Department	Date:	10-23-14
Approved By:	 Operations Department	Date:	10-23-14
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

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SIMULATOR SETUP

IC-256 **Override B410 OVDI** (Block Source Range B PB will not block)

Ensure AUDIO MULTIPLIER at NI racks is set at 1k

Ensure both SR channels are displayed on NR-45 and NR-45 alarms cleared.

INITIAL CONDITIONS: Salem Unit 2 is in Mode 2 performing a Rx startup by control rods IAW S2.OP-IO.ZZ-0003. The ICRR plot at the eight-fold position predicted criticality at 35 steps, which is the same as the Estimated Critical Rod Height. Rx Engineering predicts the Rx will go critical during the next rod pull of 25 steps.

Control Bank D is at 25 steps.

INITIATING CUE: You are the Reactor Operator. Continue the startup at Step 5.3.18.D by withdrawing Control Bank D 25 steps or until the Rx is critical, then continue with procedure to raise power and stabilize the Rx at 1×10^{-8} A in the Intermediate Range.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Identifies criticality during rod pull prior to withdrawing control bank D 25 additional steps.
2. Establishes an approximately 0.25 dpm SUR.
3. Inserts control rods to establish zero or negative SUR after determination that Source Range Channel B does not Block PRIOR to the Rx automatically tripping on High SR Flux.

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide marked up single sided paper copy of S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load, Revision 9.	Reviews procedure. When ready to accept the watch, take the Simulator from FREEZE to RUN.		
*	5.3.18.D	WITHDRAW Control Banks A, B and C STOPPING at the following hold points to obtain ICRR data: • CBA Fully Withdrawn • RIL position • CBB Fully Withdrawn • CBD at 50 Steps <u>OR</u> the 8-Fold Rod Position • CBD withdrawal limited to #50 Step increments <u>OR</u> the 8-Fold Rod Position	Announces withdrawing Control Bank D 25 steps or until criticality is achieved. Withdraws control rods while closely monitoring nuclear instrumentation, and stops withdrawing Control Bank D when identification of Rx criticality is observed. The Rx will be critical before Control Bank D have been withdrawn 25 steps.		
			Note: Step 5.3.18.F is the ICRR step, and has been completed. 5.3.18.E was N/A. 5.3.18.G has already occurred.		
	5.3.18.H	<u>IF</u> P-6 (Source Range Permissive) green light energizes....	Note: P-6 will not energize at this point.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.3.19	When criticality is achieved: A. MAKE the following announcement twice on the plant PA System: "Attention all plant personnel, Unit 2 Reactor is critical." B. RECORD Criticality Date/Time data in Attachment 2, Section 3.0, and in the Control Room Narrative Log. C. RECORD Attachment 2, Section 3.0, Surveillance Results IAW the stated Acceptance Criteria.	Announces the Rx is critical. Makes "Attention all plant personnel, Unit 2 Reactor is critical." announcement two times on the Plant PA system. Cue: The Plant operator will annotate the Control Room Log. Records date/time of criticality in Attachment 2, Section 3.0 at bottom of page. Initials SAT column of section 3.0 in Attachment 2.		
			Cue: "The CRS directs you to continue the power ascension at Step 5.3.20 by withdrawing Control Bank D to establish and maintain a 0.25 decade per minute start up rate."		
*	5.3.20	CONTINUE with Reactor Power ascension, with positive stable SUR <u>NOT</u> to exceed 1.0 dpm, by adjusting Control Bank "D" as needed.	Withdraws Control Bank D to establish and maintain a stable positive startup rate of approximately 0.25 dpm SUR.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.3.21	OBSERVE source range and intermediate range levels on NIS for $\geq$ one decade overlap during performance of next step.	Observes source range and intermediate range and identifies when at least 1 decade of overlap between them has occurred.		
*	5.3.22	When P-6 (Source Range Permissive) green light is energized, as indicated on 2RP4 ½ IR channels at 10^{-10} amps: A. SELECT one pen of NR-45 to an Intermediate Range channel and the other pen to a Power Range channel. B. BLOCK Source Range High Flux Trip by depressing both "BLOCK SOURCE RANGE A" AND "BLOCK SOURCE RANGE B" pushbuttons on console. C. ENSURE Source Range Trains A&B TRIP BLOCKED blue light is illuminated on 2RP4 AND OHA E-5 SR DET VOLT TRBL alarms.	Selects one Intermediate Range channel and one Power Range channel on recorder NR-45. Depresses BLOCK SOURCE RANGE A" AND "BLOCK SOURCE RANGE B" pushbuttons on console, and observes BLOCK SOURCE RANGE B light does NOT illuminate. Inserts control bank D to stop the power ascension and establishes a zero or negative SUR.		
*	5.3.22.D	IF Source Range Detector fails to Block, THEN INSERT Control Rods to stabilize Reactor Power:	Inserts control bank D to stop the power ascension and establishes a zero or negative SUR.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Terminate the JPM when a zero or negative SUR is established, or after the Rx trips on High Source Range Flux at 1E^5 cpm in the Source Range.		

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JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- 1 1. Task description and number, JPM description and number are identified.
- 2 2. Knowledge and Abilities (K/A) references are included.
- 3 3. Performance location specified. (in-plant, control room, or simulator)
- 4 4. Initial setup conditions are identified.
- 5 5. Initiating and terminating Cues are properly identified.
- 6 6. Task standards identified and verified by SME review.
- 7 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 8 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 39 Date 10-15-14
- 9 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: OMLOR

Date: 10-15-14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

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